Lecture 13: Comparison Tests

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

Learning Outcome

At the end of this lecture, you should be able to:

- Use the direct comparison test to tell whether a series converges or diverges when possible and know the limitations of the test.
- Use the limit comparison test to tell whether a series converges or diverges when possible and know the limitations of the test.
- Identify the series to be used for the comparison test especially when the question does not suggest any.

Direct Comparison Test

Let $\sum a_n$ and $\sum b_n$ be two series with $0 \leq a_n \leq b_n$ for all n. Then

1. If $\sum b_n$ converges, then $\sum a_n$ converges.
2. If $\sum a_n$ diverges, then $\sum b_n$ diverges.

Direct Comparison Test

Example: The series $\sum_{n=1}^{\infty} \frac{5}{5n-1}$ diverges.

Direct Comparison Test

Example: The series $\sum_{n=0}^{\infty} \frac{1}{n!}$ converges.

Direct Comparison Test

Example: The following series converges.

$$5+\frac{2}{3}+\frac{1}{7}+1+\frac{1}{2+\sqrt{1}}+\frac{1}{4+\sqrt{2}}+\frac{1}{8+\sqrt{3}}+\cdots+\frac{1}{2^n+\sqrt{n}}+\cdots$$

Direct Comparison Test

Example: Does $\sum_{n=1}^{\infty} \frac{n+1}{n^2}$ converge or diverge?

Limit Comparison Test

Suppose that $a_n > 0$ and $b_n > 0$ for all $n \geq N$, where N is an integer.

1. If $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ and $c > 0$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge.
2. If $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.
3. If $\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Limit Comparison Test

Example: $\sum_{n=1}^{\infty} \frac{2n+1}{(n+1)^2}$

Limit Comparison Test

Example: $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$

Limit Comparison Test

Example: $\sum_{n=2}^{\infty} \frac{1 + n \ln n}{n^2 + 5}$

Limit Comparison Test

Example: $\sum_{n=1}^{\infty} \frac{\ln n}{n^{\frac{3}{2}}}$

Limit Comparison Test

Example: $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)}$